HIGHWAY LANE DETECTION AND CLASSIFICATION USING VEHICLE MOTION TRAJECTORIES

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Abstract

This paper describes a clustering technique for detecting and classifying highway lanes using only vehicle motion trajectories. The detected lanes are described by low order polynomials and a directional indicator combined with simple distance metrics which permits classification of lanes into one of the following sets: *primary*, *entry*, *exit* or *secondary*. The approach taken is viewpoint independent and can be successfully applied to uncalibrated pan-tilt-zoom cameras normally used in traffic surveillance. The algorithm does not require any *a priori* road feature extraction on static images. We show that for each viewpoint and classified lane, the vehicle density and lane changes can then be analysed. The clustering technique adopted here is shown to perform better than a standard histogram lane finding approach. Experimental results are presented that show the application of this approach to multiple views obtained by a PTZ camera monitoring the junction of two intersecting highways.

Key Words

vehicle tracking, motion trajectory, lane detection, scene interpretation

1. Introduction

As the volume of traffic on roads and highways continues to grow, there is an urgent need to develop intelligent traffic surveillance systems capable of monitoring the busiest roads and junctions. The purpose of such road management systems is to provide statistical data on traffic activity (e.g. vehicle density) and to signal potentially dangerous situations.

This paper addresses the problem of lane detection and classification using standard installation traffic surveillance cameras. These non-stationary cameras are usually of the pan-tilt-zoom (PTZ) variety and are operator-controlled. They are capable of covering larger portions of the road compared to static cameras. Lane detection and classification is an important first step in building a semantic description of the scene. From this we can generate statistical data on traffic activity such as vehicle density, lane changes and the detection of dangerous/abnormal situations.

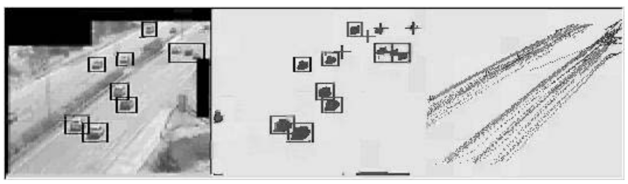
The paper is organised as follows: In section 2 we review some previous work mainly based on output from static cameras. Section 3 presents the clustering-based technique and its application to motion trajectories in some detail. Low order polynomials are used to model trajectories and RANSAC can be employed to remove outliers including lane changes. Modified K-means is used to perform clustering and we present some simple metrics to aid in lane classification. Preliminary results on trajectory clustering are given in section 4 and we present a non-trivial example of semantic scene description of a highway junction built from multiple views obtained with an ordinary operator-controlled PTZ camera. The paper concludes with a discussion and summary in section 5.

2. Review of Literature

Previous work on this topic has been based on stationary cameras which are manually calibrated and produce statistical traffic data such as vehicle speed, density and numbers of lane changes. Noteworthy contributions include higher level traffic analysis systems developed specifically for accident detection at road intersections [1][2], semantic scene description [3], mean vehicle speed estimation [4][5], video indexing and content retrieval [6]. More general techniques for object path detection and classification have also been proposed [7][8][9].

We attempt to achieve a higher level road description than in [4][5] through processing vehicle trajectories from uncalibrated views. Our work is partly inspired by [6] in which trajectories are represented by low degree polynomials. However, we choose to cluster these trajectories in order to build a model of the lane geometry using different metrics than in [2]. The only *a priori* information needed is an estimate of the lane width in image coordinates to disambiguate the zoom level used. Trajectory clustering and classification is applied in [3] to provide a natural language description of vehicle object activity in a scene, e.g. vehicle turns left at junction with low speed. We are looking for a higher level semantic description which can be applied to the overall highway lane geometry.

In this paper, we build on previous work [10] that processes video sequences from a stationary camera and generates accurate vehicle motion trajectories. It uses a Gaussian model and Adaptive Smoothness [11] to build a background model. Kalman Filter is used to predict object position when vehicle tracking is lost. Overlapping foreground objects (detected by a temporal constraint) are discarded in the analysis, since the centroid does not represent a real object position. The system output is illustrated in Fig. 1.



(a) (b) (c)

Fig. 1. *(a) Detected vehicles. (b) Foreground object labelling after background subtraction. (c) Vehicle motion trajectories generated by tracking algorithm.*

Fig. 1(a) shows the original image with a detected foreground object. In Fig.1(b) each detected and labelled blob has been assigned to a bounding box. Overlapping objects are denoted by a box with a cross in the centre. When the tracking of a particular vehicle is lost, the predicted position is calculated using a simple Kalman Filter denoted by a cross on its own. Fig.1(c) illustrates the object trajectories generated by the tracking algorithm.

These single object trajectories serve as the input to our proposed lane detection and classification system. Each trajectory comprises a set of labelled points which correspond to the position of a unique vehicle object centroid tracked through successive frames.

3. Lane Detection and Classification Method

3.1 Lane Detection by Clustering Vehicle Trajectories

In previous studies lane detection has been undertaken using feature extraction on static images, e.g. by use of Hough Transform. Whilst this might be feasible using stationary cameras in conditions of good visibility and low traffic density, it is difficult to achieve in practice. This is particularly true when using operator-controlled uncalibrated PTZ cameras and traffic density is high or image quality is poor. In general, motion data is more robust to light variation and sensor noise than spatially extracted features.

Using motion trajectories does not require *a priori* knowledge of the number of lanes or road geometry, e.g. whether a particular section of highway is straight or curved. However, the method assumes that the average lane width in image coordinates is known in advance to disambiguate the camera zoom level.

The trajectories are pre-processed to remove obvious inconsistencies in the data caused by tracking errors, sensor noise or camera motion instabilities (high winds can be a problem). Excluded trajectories include those having inter-point separation greater than some lower bound or curve length below some threshold value. By analysing all the trajectories generated, we can discover the main direction of traffic flow and determine the dominant coordinate that is monotonically increasing or decreasing. In Fig. 1(c), both x or y are admissible. This can be performed by aggregating the slopes from the start and end points of each trajectory. A similar result can be obtained using principal coordinate analysis on the tracked points. This is also used to determine which direction of the road the camera is pointing at.

We then fit a least squares polynomial of degree M for each trajectory in the chosen coordinate direction. Starting from $M = 1$, we use the average residual error of fit to ascertain the optimal value of M. If the error is greater than some threshold, M is increased by 1 and the trajectory is re-fitted ($M \leq 3$). For all the highway scenes tested, we have found that low degree polynomials up to order 3 are sufficient to describe the lane curvature and trajectories denoting lane changes. Finally, the RANSAC algorithm [12] is used in conjunction with least squares to eliminate outliers such as lane changes or noisy tracker points. Lane change trajectories must be removed in order that detection of lane centres is not biased.

Next, we apply a robust K-means clustering algorithm that works in the coefficient space of the polynomials thus greatly improving the speed and efficiency. The method used for building an initial set of clusters is described in section 3.3. An upper bound is set on the maximum number of trajectories that can belong to each cluster. The clusters centres approximately correspond to detected lanes. We propose a set of inter-trajectory distance metrics in the next section which are used to build a lane classification model.

3.2 Trajectory Distance Metrics

In [3] the separation between trajectories is calculated using the Hausdorff distance. Although this works with arbitrary points sets, it is normally expensive to compute since it involves $O(MN)$ operations, where M, N are the sizes of the trajectory point sets. This can be reduced considerably if we use the polynomial models for each trajectory to evaluate simpler expressions for the metrics. We propose to use the mean, maximum and minimum distance expressed in terms of a difference polynomial.

Let T_1 and T_2 denote two sets, where

$$\begin{aligned} T_1 &= \{ t_1(x) : x \in [a,b] \} \\ T_2 &= \{ t_2(x) : x \in [c,d] \} \end{aligned} \tag{1}$$

and t_1(x), t_2(x) are polynomial trajectory models. The difference polynomial $d(x)$ can be defined as the set D such that

$$D = \{ d(x) = t_1(x) - t_2(x) : x \in [a,b] \cup [c,d] \} \tag{2}$$

We define a set of distance measures $d(T_1,T_2)$ between T_1 and T_2 using D as follows:

$$d_{\max}(T_1,T_2) = \sup_{x\in I} \{d(x)\} \tag{3}$$

$$d_{\min}(T_1,T_2) = \inf_{x\in I} \{d(x)\} \tag{4}$$

$$d_{mean}(T_1,T_2) = \frac{1}{l_2 - l_1} \int_{l_1}^{l_2} \left(d(x)\right) dx \tag{5}$$

where l_1, l_2 are the lower and upper bounds on I, and $I = [a, b] \cup [c, d]$. In certain cases, a closed form expression can be obtained for eqs.(3) and (4) by differentiation and finding the stationary points of $d(x)$. When this fails, maxima and minima can be found by a line search technique. A geometric interpretation of $d_{\max}$ and $d_{\min}$ are shown in Fig. 2.

3.3 Modified K-means Clustering Algorithm using Distance Metrics

The following steps are used to build an initial set of trajectory clusters:

1. Create a reference linear trajectory t_R at the edge of the image parallel to the main orientation of traffic flow.
2. Build a set with a trajectory t_J that is most distant from t_R using eq. (5) and which is greater than some threshold. A sensible choice of threshold is the estimated lane width.
3. Find the next trajectory t_{J+1} ($_J$ =1,2,...) that is most distant from all trajectories present in the set and has length greater than a certain threshold.
4. If the trajectory found has a distance to the nearest neighbouring trajectory less than the estimated lane width, discard that trajectory.
5. Repeat from step 3 until no further trajectories can be added.

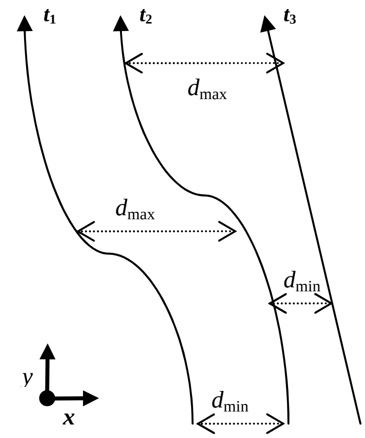


Fig. 2. *Interpretation of distance measures* $d_{\max}$, $d_{\min}$ *in terms of difference polynomial d.*

We now describe how to perform the clustering step. The trajectory-to-cluster distance is measured using eq.(3). When a trajectory is merged with a cluster, it is carried out merging the original cluster points and trajectory. In order to avoid a cluster becoming biased by merger with an outlier trajectory, each cluster is only remodelled using RANSAC for every 5 trajectories considered.

For all the trajectories, we measure the trajectory-to-cluster distance and merge each trajectory t to the nearest cluster if $d_{\max} < \tau$ (for some threshold τ) and curve length $|t| > L_{\min}$. For all t_J, we calculate min($d_{\min}$'s) and add the trajectory indexed by this minimum to the current cluster. The cluster means are updated after a fixed number of repetitions. The number of trajectories added to each cluster are limited to 20 and for every 5 trajectories added, RANSAC is applied to modelling the cluster in order to eliminate outliers. It is found that use of $d_{\max}$ for measuring trajectory-to-cluster separation discards most trajectories representing lane changes. RANSAC then removes most remaining outliers erroneously added to the cluster.

3.4 Lane Classification

The camera orientation is established by calculating a linear fit to the main direction of traffic flow. This gives the approximate angle between the main highway and the direction in which the camera is pointing. The direction of traffic flow is classified as (*right* $\vee$ *left*). In Fig. 1, the angle is about 45^0 and the main direction flow is *right* (i.e. vehicles travelling from left to right).
The main trajectory direction is classified as (*positive* $\vee$ *negative*) by inspecting the dominant values of (x, y) coordinate changes (Δx, Δy). If positive values of Δx predominate, the direction is classified as *positive*.

The trajectory clusters representing lanes are classified into one of the following (*primary* $\vee$ *entry* $\vee$ *exit* $\vee$ *secondary*). For the sake of simplicity, we assume that every highway has at least four primary lanes, two in each direction of flow. In abnormal situations such as temporary roadworks and contra flow, this assumption may not hold. The four primary lanes in the centre of the highway are found by searching for neighbouring lanes having opposite direction flow. The distance measures are initially applied to the first two lanes having the same direction.

We use l to denote a particular lane and L to represent the set of all lanes found in the scene. The lane classification scheme can be expressed as follow:

$\forall l \in L$,
if
$$d_{\max}(l_i, l_{i-1}) \le \delta \wedge d_{\min}(l_i, l_{i-1}) \le \delta \Rightarrow (primary)$$
else
$$\text{if } d_{\max}(l_i, l_{i-1}) > \delta \wedge d_{\min}(l_i, l_{i-1}) > \delta \wedge d_{\text{mean}}(l_i, l_{i-1}) > \delta \Rightarrow (secondary)$$
else
$$(entry \vee exit)$$

The choice of entry or exit is determined by the angle this lane makes with the previously classified lane and if the direction of traffic flow is positive or negative.

3.5 Traffic Flow Analysis

The approximate number of vehicles per lane can be easily counted using the trajectory data. Obviously this excludes overlapping vehicles that are discarded in the tracking phase. Using a previously classified lane as a query, and assuming that the lanes are numbered in spatial sequence, a set Ω is built containing all trajectories which are closer to the query lane (using $d_{\max}$) than are to the neighbouring lanes (using $d_{\min}$). All lane changes are discarded since maximum distance to query lane is greater then minimum distance. The set Ω contains only those vehicle trajectories exclusively travelling in lane l. In the following, L is the set of lanes and T the set of trajectories.

$$\forall l \in L, \forall t \in T$$
$$d_{\max}(l_i, t) < d_{\min}(l_{i-1}, t) \wedge d_{\max}(l_i, t) < d_{\min}(l_{i+1}, t) \Rightarrow t \in \Omega$$

The set of vehicle trajectories representing lane changes can be inferred from the above scheme by the principle of mutual exclusivity. That is, by excluding all the trajectories $t \in \Omega$ and retaining the others, i.e. a trajectory represents either a vehicle that keeps to the same lane or changes lane.

4. Experimental Results

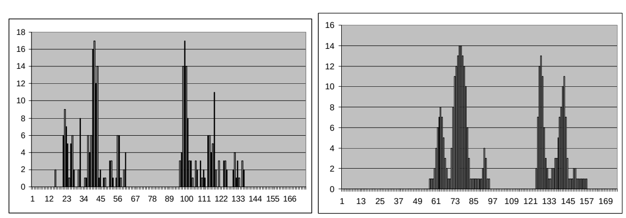
The algorithm was tested on two different highway surveillance sequences. The first was recorded with a stationary camera having a resolution of 176x288 pixels and capture rate of 15 frames/sec. The second sequence was taken by a higher resolution camera (352x288) at 25 frames/sec. Unfortunately, the recording of the second sequence was made on a very windy day and apparent uncontrolled camera motion often resulted in poor tracking results and fragmented trajectories. Nevertheless, this presents us with a challenging data set. In the first sequence, however, the tracking module generates satisfactory results.

4.1 Lane Detection

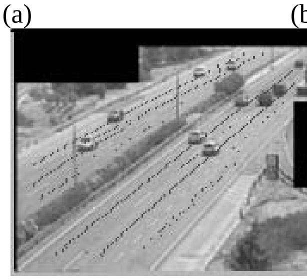
In order to compare our proposed technique with accumulated histograms (or motion activity maps) used in [5], we process lane detection results using this approach. In Fig. 3, we depict the original scene and the accumulated single vehicle trajectories. The lane detection stage then proceeds by blurring the histogram peaks and plotting the lane centres. It can be observed in Fig. 4 that some lanes have a higher vehicle count than others and that this affects the accuracy of lane detection. The solution is very inaccurate for lanes with less traffic. In may cases lane changes are perceived to denote a new lane.

(a) (b)

Fig. 3. *(a) Original scene. (b) Accumulated trajectory histograms.*



(a) (b)



(c)

Fig. 4. *(a) Histogram of detected vehicles. (b) Result after applying blurring operator. The x-axis denotes the x-pixel coordinate and the y-axis the number of vehicles. (c) Lane detection result.*

The results of applying the initial clustering algorithm described in section 3.3 are shown in Fig. 5. Initial clusters are represented by solid lines, whereas the original trajectories of single vehicles are depicted by dots. All the initial clusters lie on the edges of the point clouds since it attempts to maximise the $d_{\max}$ inter-cluster distance.

Fig. 5. *Results of initial clustering algorithm for the two camera views. The solid lines denote cluster centres and dots represent individual vehicle trajectories.*

The final clusters produced after termination of the algorithm are illustrated in Fig. 6. The values indicate the numbers of trajectories added to the initial clusters for each lane. Fig. 6(a) and 6(b) show the result with and without applying RANSAC. The RANSAC result is slightly better on the inner primary highway furthest from the camera due to outlier elimination.

(a) (b)

Fig. 6. *Final results of detecting lane cluster centres (a) using RANSAC approach,(b) without applying RANSAC.*

In comparison with Fig. 3(c), it can be seen that our algorithm gives more stable lane detection results than histogram accumulation. For incomplete trajectories, the clustering algorithm works fairly well when the trajectories can be adequately modelled by linear fit. It is robust to lane changes and missing data. As the polynomial degree increases (typically required for curved lanes), the algorithm tends to become unstable.

4.2 Traffic Analysis and Lane Classification

We applied the lane classification scheme to a relatively complex highway scene near Lisbon, Portugal. A single PTZ camera observes two intersecting highways, the junction of ‘A5’ and ‘CRIL’. CRIL has 4 primary lanes, 2 entry lanes and 1 exit lane in the field of view closest to the camera, whereas the A5 has 7 primary lanes and 4 exit lanes. The traffic is usually heaviest on the A5. In this case, the sequences were recorded on a very windy day resulting in unwanted camera motions. This motion can reach a maximum of 5 pixels and represents a challenging test scenario. Typical camera views and a schematic of the road layout are shown in Figs. 7 and 8. Main directions and lane numbers are summarised in Table 1.

(a) (b) (c) (d)

Fig. 7. *PTZ camera views for each of the four main directions of traffic flow. (a) and (b) highway ‘CRIL’, (c) and (d) highway ‘A5’.*

Highway name	**A5**	**CRIL**
Minimum Number of Lanes	11	6
Direction Looking Right	Cascais	Amadora
Direction Looking Left	Lisbon	Queijas

Table 1. *Summary of total numbers of lanes and destinations for main directions of traffic flow.*

The camera orientation is easily found by measuring the angle of main direction flow with the road.

To determine which highway is currently being viewed, we use the number of detected lanes together with the information presented in Table 1. After the camera orientation is found, the destination IDs can also be established from the table. This is the only calibration information required to perform the lane classification once the lane detection is completed. The classification of entry or exit lanes can be retrieved from the trajectory data.

The algorithm performs well when the lanes are nearly straight, e.g. CRIL highway. Here, the number of vehicles in each lane/direction and number of lane changes can be successfully calculated. This information is very useful for automated traffic flow analysis. The results are shown in Fig. 9.

The results obtained for the A5 highway, shown in Fig. 10, are not as good, especially when the camera is looking right. There are several reasons for this. A greater number of lanes means that vehicle resolution is smaller, and hence vehicle detection is more sensitive to camera motion. Also when the camera is looking right, the zoom level is greater than when looking left and this produces a degraded result. The success of the classification scheme is highly dependent on the quality of the lane modelling, which in turn is dependent on the output of vehicle tracking module. Hence, the results tend to be better for straight roads.

Further results of detection and classification of lanes with colour images can be found at http://omni.isr.ist.utl.pt/~jpqm/inteltraf.htm.

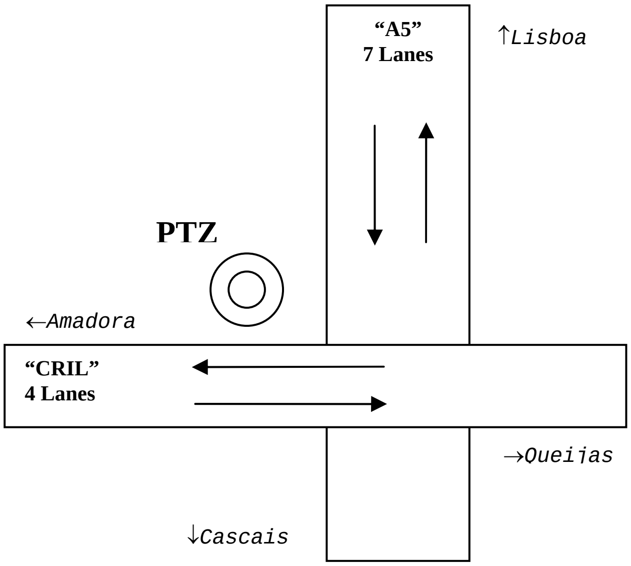


Fig. 8. *Schematic layout for A5/CRIL highway intersection.*

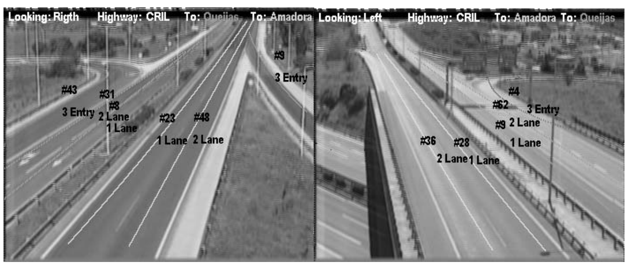


Fig. 9. *Lane detection and classification result for highway CRIL. The numbers of vehicles are also shown.*

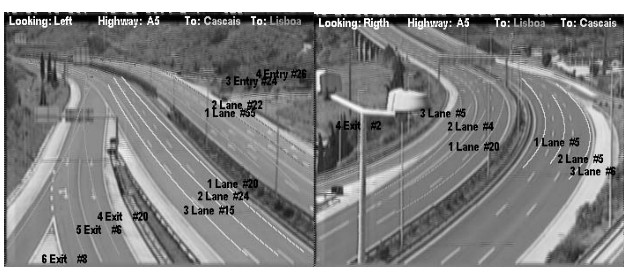


Fig. 10. *Lane detection and classification result for highway A5. Vehicle numbers in each lane are also shown.*

5. Discussion and Conclusions

We have presented an algorithm which performs lane detection and classification for highway scenes using only vehicle trajectory data and a basic knowledge of the road layout. This can be used to generate a higher level scene description which is useful for automated traffic flow analysis. The method works with both uncalibrated PTZ and static road traffic cameras and can cope with variable lane geometry. The lane detection stage is robust to vehicle lane changes and missing or incomplete trajectory data through use of RANSAC model fitting.

We have demonstrated an improvement in lane detection results using our algorithm over the standard accumulated histogram (or activity map) approach. The classification scheme works well in the case of straight (or almost straight) roads but performs poorly when road curvature is high. This could be due to instability in vehicle tracking leading to incomplete trajectories, or unsuitable distance metrics. In further work we aim to explore more appropriate clustering metrics and refine the tracking and classification algorithms.

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